Empathy in Real vs. Synthetic Clinician Comments (Pilot 2)

0.1 Summary

Across 3 synthetic models (Gemini, Grok, Openai) and 3 prompting methods (DG, SAG, SAGII), real comments generally show significantly higher empathy ratings across all three mechanisms (Emotional Reactions, Explorations, Interpretations), with one exception: for Interpretations, Grok scores significantly *higher* than real across all three prompting strategies (DG/SAG/SAGII).

0.2 Data

3000 Reddit posts were used, each paired with its real clinician comment(s) and synthetic comments from 3 models x 3 prompting methods. 62161 comments were classified in total: 3882 real and 58279 synthetic (6475 per model/method group on average). Predicted empathy comes from the same three fine-tuned RoBERTa bi-encoder classifiers used in pilot 1 (Emotional Reactions, Interpretations, Explorations; see `../2_training`), each producing an ordinal label: 0 = none, 1 = weak, 2 = strong.

0.3 Mean predicted empathy by group

Empathy labels are ordinal (0 = none, 1 = weak, 2 = strong). The means shown below implicitly treat the labels as equally spaced, and are shown only as a quick, monotonic summary (Fig. 1). The actual label distribution (the ordinal-honest view) is plotted underneath it as proportions per group (Fig. 2).

0.4 Real vs. synthetic (Mann-Whitney U, Benjamini-Hochberg adjusted)

Table 2 shows that all real-vs-synthetic contrasts are significant, such that real ones score significantly higher. Notable exceptions, where the synthetic ones score higher are marked in red.

0.5 Mixed-effects models

The Mann-Whitney tests above treat every comment as independent, but they aren't: several synthetic comments (and sometimes several real ones) come from the *same* Reddit thread, so they share whatever makes that particular topic more or less emotionally charged or explorable. To account for that, we run two mixed-effects models per mechanism:

- **A - omnibus** (`label ~ group + log(length) + (1|post_id)`, all 62161 comments): tests whether the 10-level group factor (real + 9 model/method combinations) matters at all.
- **B - model × method** (`label ~ model * method + log(length) + (1|post_id)`, synthetic comments only - "real" has no model/method to cross): tests the model main effect, the method main effect, and whether they interact.

Emotional Reactions uses all 3 empathy levels, so it's fit as a cumulative link mixed model (`ordinal::clmm`). **Interpretations** and **Explorations** never predict label 1 anywhere in this data (0 occurrences out of 62161 comments, checked directly), so a 3-level ordinal model would be degenerate for them—they're fit instead as binomial mixed models (`lme4::glmer`) on P(label = 2).

Emotional Reactions's models did not fully converge (max|grad| = 0.00905 (didn't fully settle) for model A, max|grad| = 0.00131 (didn't fully settle) for model B). Both gradients are small (no diverging coefficients, no crash), consistent with the known separation risk in `grok_sagii` (0/6475 comments at label 2 for that combination) rather than a broken model. Accordingly, ER's results below are worth treating as directionally informative rather than fully settled; IP and EX converged cleanly.

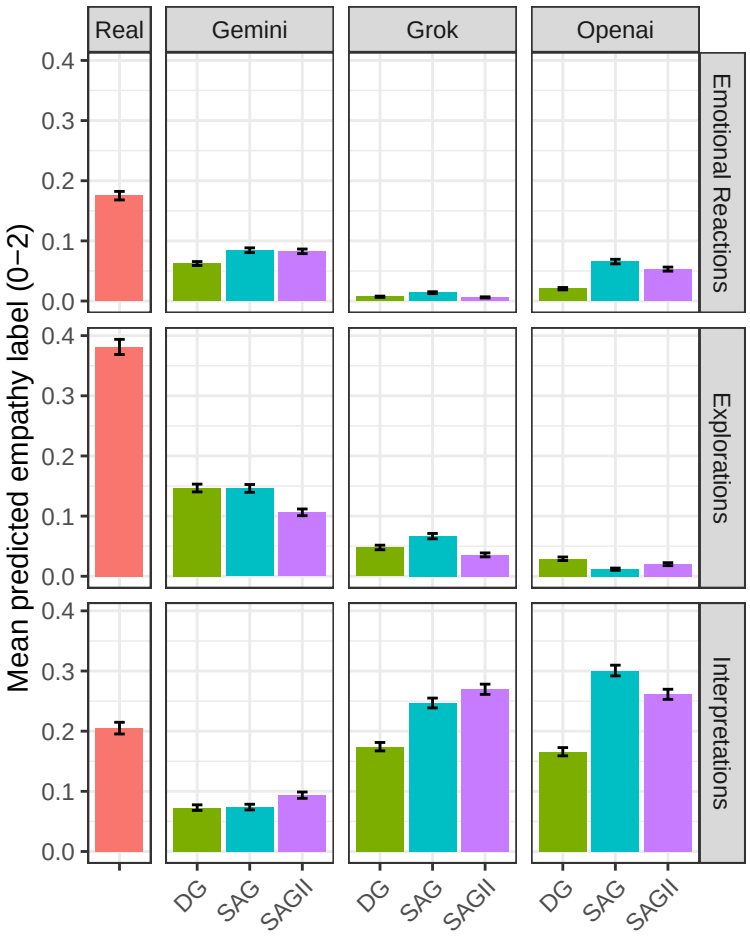


Figure 1: Average predicted empathy per population.

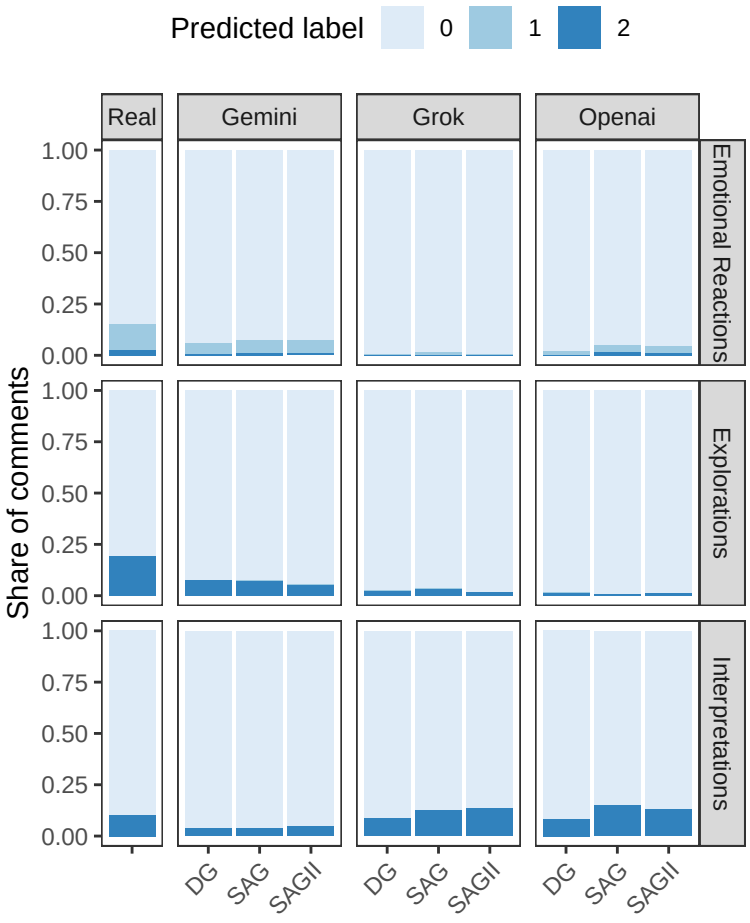


Figure 2: Stacked predicted label proportions per population.

Table 1: Predicted label proportions by group (same data as Fig. 2, as a table) - useful for spotting cells too sparse to trust, e.g. where P(label=2) is close to 0.

Mechanism	Model	Method	n	P(label=0)	P(label=1)	P(label=2)
Emotional Reactions	Real		3882	0.850	0.124	0.026
	Gemini	DG	6476	0.943	0.053	0.005
		SAG	6476	0.927	0.061	0.012
		SAGII	6474	0.927	0.063	0.010
	Grok	DG	6475	0.993	0.007	0.000
		SAG	6475	0.987	0.012	0.001
		SAGII	6475	0.994	0.006	0.000
	Openai	DG	6476	0.982	0.015	0.003
		SAG	6476	0.950	0.034	0.016
		SAGII	6476	0.958	0.031	0.011
Explorations	Real		3882	0.809	0.000	0.191
	Gemini	DG	6476	0.927	0.000	0.073
		SAG	6476	0.927	0.000	0.073
		SAGII	6474	0.947	0.000	0.053
	Grok	DG	6475	0.976	0.000	0.024
		SAG	6475	0.967	0.000	0.033
		SAGII	6475	0.982	0.000	0.018
	Openai	DG	6476	0.985	0.000	0.015
		SAG	6476	0.994	0.000	0.006
		SAGII	6476	0.990	0.000	0.010
Interpretations	Real		3882	0.897	0.000	0.103
	Gemini	DG	6476	0.964	0.000	0.036
		SAG	6476	0.963	0.000	0.037
		SAGII	6474	0.953	0.000	0.047
	Grok	DG	6475	0.913	0.000	0.087
		SAG	6475	0.877	0.000	0.123
		SAGII	6475	0.865	0.000	0.135
	Openai	DG	6476	0.917	0.000	0.083
		SAG	6476	0.850	0.000	0.150
		SAGII	6476	0.869	0.000	0.131

Table 2: Mann-Whitney U test.

Mechanism	Model	Method	Mean (real)	Mean (synthetic)	p	p (BH-adj.)	Rank-biserial r
Emotional Reactions	Gemini	DG	0.175	0.062	0.0000	0.0000	0.093
		SAG	0.175	0.084	0.0000	0.0000	0.077
		SAGII	0.175	0.083	0.0000	0.0000	0.077
	Grok	DG	0.175	0.007	0.0000	0.0000	0.143
		SAG	0.175	0.014	0.0000	0.0000	0.137
		SAGII	0.175	0.006	0.0000	0.0000	0.144
	Openai	DG	0.175	0.021	0.0000	0.0000	0.132
		SAG	0.175	0.066	0.0000	0.0000	0.099
		SAGII	0.175	0.053	0.0000	0.0000	0.107
Explorations	Gemini	DG	0.381	0.147	0.0000	0.0000	0.117
		SAG	0.381	0.146	0.0000	0.0000	0.118
		SAGII	0.381	0.106	0.0000	0.0000	0.137
	Grok	DG	0.381	0.048	0.0000	0.0000	0.167
		SAG	0.381	0.067	0.0000	0.0000	0.157
		SAGII	0.381	0.036	0.0000	0.0000	0.173
	Openai	DG	0.381	0.029	0.0000	0.0000	0.176
		SAG	0.381	0.012	0.0000	0.0000	0.185
		SAGII	0.381	0.020	0.0000	0.0000	0.181
Interpretations	Gemini	DG	0.205	0.073	0.0000	0.0000	0.066
		SAG	0.205	0.074	0.0000	0.0000	0.066
		SAGII	0.205	0.094	0.0000	0.0000	0.056
	Grok	DG	0.205	0.174	0.0089	0.0089	0.015
		SAG	0.205	0.247	0.0013	0.0013	-0.021
		SAGII	0.205	0.270	0.0000	0.0000	-0.032
	Openai	DG	0.205	0.166	0.0008	0.0008	0.020
		SAG	0.205	0.301	0.0000	0.0000	-0.048
		SAGII	0.205	0.261	0.0000	0.0000	-0.028

Table 3: Convergence status for each fitted model.

Mechanism	Model	Type	Converged	Note
Emotional Reactions	A_group_omnibus	clmm	FALSE	max|grad| = 0.00905 (didn't fully settle)
Emotional Reactions	B_model_x_method	clmm	FALSE	max|grad| = 0.00131 (didn't fully settle)
Interpretations	A_group_omnibus	glmer	TRUE	NA
Interpretations	B_model_x_method	glmer	TRUE	NA
Explorations	A_group_omnibus	glmer	TRUE	NA
Explorations	B_model_x_method	glmer	TRUE	NA

Table 4: Model A: omnibus test for group (real + 9 model/method combinations), replacing Kruskal-Wallis.

Mechanism	Chi-sq	df	p
Emotional Reactions	1744.8	9	0
Interpretations	1241.7	9	0
Explorations	2035.2	9	0

Table 5: Model A: each (model, method) group vs. real (OR below 1 = lower odds of higher empathy than real; rows with OR above 1 shaded).

Mechanism	Group	Odds ratio	OR 95% LCL	OR 95% UCL	p (BH-adj.)
Emotional Reactions	gemini_dg	0.272	0.232	0.320	0
Emotional Reactions	openai_dg	0.050	0.040	0.062	0
Emotional Reactions	grok_dg	0.027	0.019	0.037	0
Emotional Reactions	gemini_sag	0.355	0.305	0.412	0
Emotional Reactions	openai_sag	0.137	0.115	0.163	0
Emotional Reactions	grok_sag	0.045	0.035	0.057	0
Emotional Reactions	gemini_sagii	0.333	0.286	0.386	0
Emotional Reactions	openai_sagii	0.109	0.091	0.131	0
Emotional Reactions	grok_sagii	0.020	0.014	0.028	0
Interpretations	gemini_dg	0.628	0.514	0.767	0
Interpretations	openai_dg	0.612	0.522	0.718	0
Interpretations	grok_dg	2.106	1.760	2.520	0
Interpretations	gemini_sag	0.555	0.455	0.676	0
Interpretations	openai_sag	0.693	0.596	0.805	0
Interpretations	grok_sag	2.213	1.877	2.608	0
Interpretations	gemini_sagii	0.599	0.498	0.721	0
Interpretations	openai_sagii	0.580	0.498	0.675	0
Interpretations	grok_sagii	2.835	2.401	3.349	0
Explorations	gemini_dg	0.185	0.159	0.215	0
Explorations	openai_dg	0.032	0.025	0.041	0
Explorations	grok_dg	0.044	0.036	0.054	0
Explorations	gemini_sag	0.193	0.166	0.224	0
Explorations	openai_sag	0.015	0.011	0.022	0
Explorations	grok_sag	0.073	0.060	0.087	0
Explorations	gemini_sagii	0.134	0.114	0.157	0
Explorations	openai_sagii	0.027	0.020	0.036	0
Explorations	grok_sagii	0.034	0.028	0.043	0

The group effect is significant for all three mechanisms (Table 4). Comment length (log-transformed) is included as a covariate in both models and is a significant predictor everywhere (Model A: all three $p < 0.001$), but not in the same direction: Emotional Reactions (longer), Interpretations (longer) score higher with more words, while Explorations scores *lower* with more words—plausibly because exploratory, question-asking replies tend to be short follow-ups rather than long ones.

Table 5: real scores significantly higher than every synthetic group for Emotional Reactions and Explorations. Interpretations is the exception—Grok scores significantly *higher* than real for all three prompting strategies, while Gemini and OpenAI still score lower.

This Grok/OpenAI split isn't what the raw Mann-Whitney table shows (there, `grok_dg` was lower and `openai_sag`/`openai_sagii` were higher—the opposite pattern). Refitting without `log(length)` reproduces the raw pattern almost exactly, so this looks like a length confound: OpenAI writes much longer comments than Grok, and length strongly predicts this label, so part of OpenAI's raw advantage—and part of Grok's raw shortfall with DG—is just comment length.

The model x method interaction is significant for all three mechanisms, i.e. the effect of a prompting method genuinely depends on which model generated it. The one main effect that drops out is **Interpretations's method effect** ($p = 0.45$): once model is accounted for, which prompting method was used doesn't matter on its own for Interpretations.

Since the interaction is significant everywhere, pooled main-effect means (averaged over the other factor) would be misleading here - e.g. a model's "average" effect can look muted if it does well with one method

Table 6: Model B: model, method, and model x method tests (synthetic comments only), replacing the pooled by-method/by-model comparisons.

Mechanism	Term	Chi-sq	df	p
Emotional Reactions	model	865.8	2	0.0000
Emotional Reactions	method	50.2	2	0.0000
Emotional Reactions	model:method	37.7	4	0.0000
Interpretations	model	1278.7	2	0.0000
Interpretations	method	1.6	2	0.4529
Interpretations	model:method	41.8	4	0.0000
Explorations	model	591.4	2	0.0000
Explorations	method	49.0	2	0.0000
Explorations	model:method	52.4	4	0.0000

and poorly with another. Instead, two sets of simple-effects contrasts: models compared within each method, and methods compared within each model.

Table 7: Model B: pairwise model contrasts within each method.

Mechanism	Method	Contrast	Odds ratio	OR 95% LCL	OR 95% UCL	p (BH-adj.)
Emotional Reactions	dg	gemini / openai	5.953	4.603	7.700	0.0000
Emotional Reactions	dg	gemini / grok	11.030	7.927	15.348	0.0000
Emotional Reactions	dg	openai / grok	1.853	1.252	2.741	0.0023
Emotional Reactions	sag	gemini / openai	2.791	2.181	3.572	0.0000
Emotional Reactions	sag	gemini / grok	8.605	6.684	11.079	0.0000
Emotional Reactions	sag	openai / grok	3.083	2.241	4.243	0.0000
Emotional Reactions	sagii	gemini / openai	3.300	2.597	4.195	0.0000
Emotional Reactions	sagii	gemini / grok	18.314	12.928	25.944	0.0000
Emotional Reactions	sagii	openai / grok	5.549	3.687	8.351	0.0000
Interpretations	dg	gemini / openai	1.424	1.178	1.721	0.0003
Interpretations	dg	gemini / grok	0.279	0.236	0.330	0.0000
Interpretations	dg	openai / grok	0.196	0.165	0.232	0.0000
Interpretations	sag	gemini / openai	1.241	1.023	1.505	0.0294
Interpretations	sag	gemini / grok	0.252	0.215	0.296	0.0000
Interpretations	sag	openai / grok	0.203	0.174	0.238	0.0000
Interpretations	sagii	gemini / openai	1.504	1.262	1.792	0.0000
Interpretations	sagii	gemini / grok	0.191	0.165	0.221	0.0000
Interpretations	sagii	openai / grok	0.127	0.108	0.150	0.0000
Explorations	dg	gemini / openai	3.830	2.968	4.943	0.0000
Explorations	dg	gemini / grok	5.918	4.783	7.323	0.0000
Explorations	dg	openai / grok	1.545	1.147	2.081	0.0045
Explorations	sag	gemini / openai	4.069	2.720	6.086	0.0000
Explorations	sag	gemini / grok	2.833	2.339	3.430	0.0000
Explorations	sag	openai / grok	0.696	0.460	1.055	0.0875
Explorations	sagii	gemini / openai	1.870	1.343	2.605	0.0003
Explorations	sagii	gemini / grok	5.461	4.288	6.955	0.0000
Explorations	sagii	openai / grok	2.920	1.987	4.289	0.0000

Table 8: Model B: pairwise method contrasts within each model.

Mechanism	Model	Contrast	Odds ratio	OR 95% LCL	OR 95% UCL	p (BH-adj.)
Emotional Reactions	gemini	dg / sag	0.762	0.649	0.894	0.0025
Emotional Reactions	gemini	dg / sagii	0.817	0.694	0.963	0.0251
Emotional Reactions	gemini	sag / sagii	1.073	0.923	1.247	0.4595
Emotional Reactions	openai	dg / sag	0.357	0.277	0.460	0.0000
Emotional Reactions	openai	dg / sagii	0.453	0.350	0.587	0.0000
Emotional Reactions	openai	sag / sagii	1.269	1.060	1.520	0.0171
Emotional Reactions	grok	dg / sag	0.594	0.405	0.873	0.0153
Emotional Reactions	grok	dg / sagii	1.357	0.867	2.126	0.2455
Emotional Reactions	grok	sag / sagii	2.284	1.534	3.402	0.0002
Interpretations	gemini	dg / sag	1.168	0.964	1.417	0.1703
Interpretations	gemini	dg / sagii	1.149	0.955	1.384	0.2012
Interpretations	gemini	sag / sagii	0.984	0.819	1.181	0.8708
Interpretations	openai	dg / sag	1.019	0.894	1.160	0.8446
Interpretations	openai	dg / sagii	1.214	1.064	1.385	0.0090
Interpretations	openai	sag / sagii	1.192	1.067	1.330	0.0048
Interpretations	grok	dg / sag	1.055	0.927	1.202	0.4903
Interpretations	grok	dg / sagii	0.786	0.693	0.892	0.0006
Interpretations	grok	sag / sagii	0.745	0.663	0.837	0.0000
Explorations	gemini	dg / sag	0.781	0.663	0.919	0.0074
Explorations	gemini	dg / sagii	0.985	0.824	1.178	0.8708
Explorations	gemini	sag / sagii	1.262	1.063	1.498	0.0153
Explorations	openai	dg / sag	0.830	0.541	1.273	0.4812
Explorations	openai	dg / sagii	0.481	0.333	0.696	0.0004
Explorations	openai	sag / sagii	0.580	0.380	0.884	0.0191
Explorations	grok	dg / sag	0.374	0.293	0.476	0.0000
Explorations	grok	dg / sagii	0.909	0.693	1.192	0.5525
Explorations	grok	sag / sagii	2.432	1.889	3.132	0.0000